German botanist Andreas Schimper published his ideas on plant geography in 1898.

Botanists such as Carl Linnaeus and Alexander von Humboldt had written about plant distributions in the 18th and early 19th centuries. The widely traveled Humboldt understood that climate was one of the key factors governing where plants did and did not grow. Schimper went one step further than Humboldt by explaining that similar vegetation types arise under similar climatic conditions in different parts of the world. He then produced a global classification of vegetation zones that reflected this observation.

Schimper's 1898 book *Plant-geography upon a Physiological Basis* ran to 870 pages and is one of the largest ecology monographs written by a single author. A synthesis of plant geography and plant physiology (the functioning of plants), it became the foundation of the study of plant ecology. Schimper explained that the connection between the structures of plants and the external conditions they faced in different places was the key to what he described as "ecological plant-geography." Vegetation was divided into broad tropical, temperate, arctic, mountain, and aquatic zones, then subdivided further, according to the prevailing climate. For example, tropical vegetation was divided into savanna, thorn-forest, woodland, tropical rain forest, or woodland with a pronounced dry season, according to whether the climate was wet all year round, seasonally wet, or mostly dry.